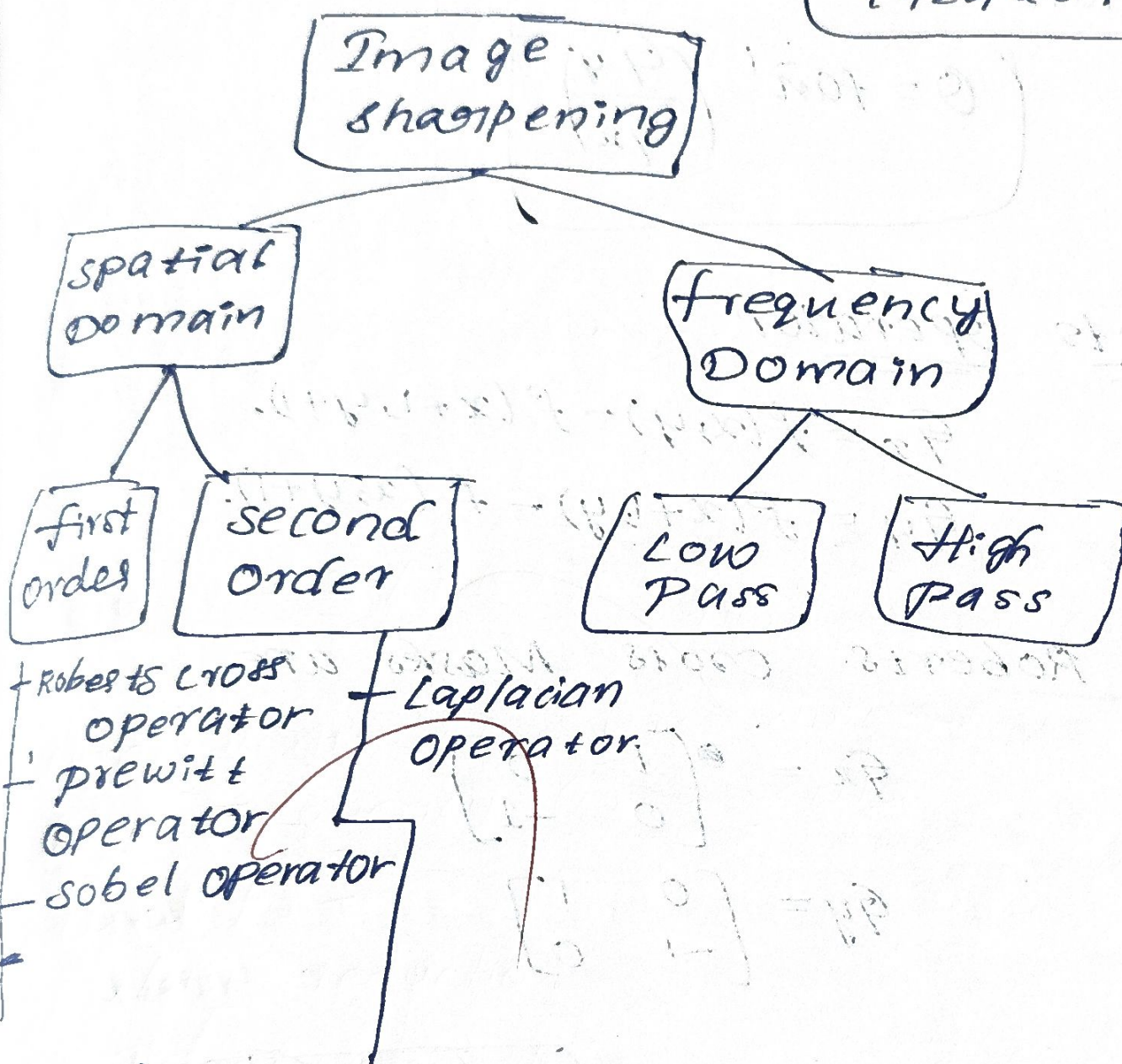


Image sharpening:-

CO2 - AT, 28/7/26

classification of Image sharpening techniques:-

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Gradient Magnitude:-

The strength of an edge is measured using the gradient magnitude

It is calculated using both Horizontal and Vertical gradients.

$$G = \sqrt{G_x^2 + G_y^2}$$

problem understanding	method selection	solution process	accuracy of calc.	Interpretation of Results	total
20	20	25	20	15	100

Gradient Direction:-

The direction of the maximum intensity change is called the Gradient direction.

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

Roberts Operator

$$G_x = f(x, y) - f(x+1, y+1)$$

$$G_y = f(x+1, y) - f(x, y+1)$$

Two Roberts cross Masks are

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

*

~~15~~ ~~20~~

~~25~~ ~~35~~

15	20	30	35
25	35	50	55
40	60	75	85
50	70	95	110

Window (or) Mask (or) Kernel

$$\text{Window (or) Mask (or) Kernel} = \begin{bmatrix} 15 & 20 \\ 25 & 35 \end{bmatrix}$$

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$Gx = 15 + 0 + 0 + (-35) = -20$$

$$Gy = 0 + 20 + (-25) + 0 = -5$$

$$= \sqrt{Gx^2 + Gy^2}$$

$$= \sqrt{(-20)^2 + (-5)^2}$$

$$= \sqrt{400 + 25}$$

$$= \sqrt{425} = 20.61$$

$$\text{window}(1,2) = \begin{bmatrix} 20 & 30 \\ 3 & 5 & 50 \end{bmatrix}$$

$$Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$Gx = 20 + 0 + 0 - 50 = -30 \quad Gy = 0 + 30 - 35 + 0 = -5$$

$$\sqrt{(-30)^2 + (-5)^2} = \sqrt{900 + 25} = \sqrt{925} = 30.4$$

$$\text{window}(1,3) = \begin{bmatrix} 30 & 35 \\ 50 & 55 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$Gx = 30 + 0 + 0 - 55 = -25 \quad Gy = 0 + 35 - 50 + 0 = -15$$

$$\Rightarrow \sqrt{(-25)^2 + (-15)^2} = \sqrt{625 + 225} = \sqrt{850} = 29.15$$

$$\text{window}(2,1) = \begin{bmatrix} 25 & 35 \\ 40 & 60 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$Gx = 25 + 0 + 0 - 60 = -35 \quad Gy = 0 + 35 - 40 + 0 = -5$$

$$\Rightarrow \sqrt{(-35)^2 + (-5)^2} = \sqrt{1225 + 25} = \sqrt{1250} = 35.35$$

$$\text{window } (2,2) = \begin{bmatrix} 35 & 50 \\ 60 & 75 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Gx = 35 + 0 + 0 - 75$$

$$= -40$$

$$Gy = 0 + 50 - 60 + 0$$

$$= -10$$

$$\sqrt{(-40)^2 + (-10)^2} = \sqrt{1600 + 100} = \sqrt{1700}$$

$$= 41.23$$

$$\text{window } (2,3) = \begin{bmatrix} 50 & 55 \\ 75 & 85 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$Gx = 50 + 0 + 0 - 85$$

$$= -35$$

$$Gy = 0 + 55 - 75 + 0$$

$$= -20$$

$$\sqrt{(-35)^2 + (-20)^2} = \sqrt{1225 + 400} = \sqrt{1625}$$

$$= 40.31$$

$$\text{window } (3,1) = \begin{bmatrix} 40 & 60 \\ 50 & 70 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 \\ -10 \end{bmatrix}$$

$$Gx = 40 + 0 + 0 - 70$$

$$= -30$$

$$Gy = 0 + 60 - 50 + 0$$

$$= 10$$

$$\Rightarrow \sqrt{(-30)^2 + 100} = \sqrt{1000} = 31.6$$

$$\text{window } (3,2) = \begin{bmatrix} 60 & 75 \\ 70 & 95 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Gy = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$Gx = 60 + 0 + 0 - 95 \\ = -35$$

$$Gy = 0 + 75 - 70 + 0 \\ = 5$$

$$\Rightarrow \sqrt{(-35)^2 + (5)^2} = \sqrt{1225 + 25} = \sqrt{1250} \\ = 35.35$$

$$\text{window } (3,3) = \begin{bmatrix} 75 & 85 \\ 95 & 110 \end{bmatrix} \quad Gx = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Gy = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$$Gx = 75 + 0 + 0 - 110 \\ = -35$$

$$Gy = 0 + 85 - 95 + 0 \\ = -10$$

$$\Rightarrow \sqrt{(1225 + 100)} = \sqrt{1325} = 36.40$$

$$\begin{bmatrix} 26.61 & 30.4 & 29.18 \\ 35.35 & 41.23 & 40.31 \\ 31.6 & 35.35 & 36.40 \end{bmatrix}$$

Prewitt operator:- (3x3 window)

$$Gx = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad Gy = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

$$* \begin{bmatrix} 25 & 40 & 55 \\ 45 & 65 & 85 \\ 70 & 95 & 120 \end{bmatrix}$$

$$Gx = -25 + 0 + 55 - 45 + 0 + 85 - 70 + 0 + 120 \\ = 120$$

$$Gy = 25 + 40 + 55 + 0 + 0 + 0 - 70 - 95 - 120 \\ = -165$$

$$\Rightarrow \sqrt{(120)^2 + (-165)^2}$$

$$= \sqrt{14400 + 27225} = \sqrt{41625}$$

$$= 204.0$$

$$G = |G_x| + |G_y|$$

$$= 120 + 165 = 285$$

Sobel operator:-

(3x3 window).

$$G_x = 3 \times 3$$

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$* \begin{bmatrix} 10 & 10 & 90 \\ 15 & 15 & 95 \\ 20 & 20 & 100 \end{bmatrix}$$

$$G_x = \underline{-10 + 0 + 90} - \underline{30 + 0 + 190} - \underline{20 + 0 + 100}$$

$$= 360 - 60 = 300$$

$$G_y = 10 + 20 + 90 + 0 + 0 + 0 - 20 - 40 - 100$$

$$= 120 - 160$$

$$= -40$$

$$\Rightarrow \sqrt{10200 + 1600} = \sqrt{2500} = 50$$

$$0.51 + 0.108 - 28 + 0 = \sqrt{104000} = 322.49$$

$$\Rightarrow 360$$

$$G = |G_x| + |G_y| = 360$$

Laplacian operator

20	30	40	50
30	45	60	75
45	60	80	95
60	80	100	120

$$L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \checkmark$$

Laplacian is a second order derivative edge detection technique. It doesn't calculate gradient direction like Sobel, Prewitt, Robert. It directly measures the rate of intensity change.

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\begin{pmatrix} 20 & 30 & 40 \\ 30 & 45 & 60 \\ 45 & 60 & 80 \end{pmatrix} \quad G_x = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$L G_x = 0 - 30 + 0 - 30 + 180 - 60 + 0 - 60 + 0 = 0. \text{ (no edge)}$$

~~G_y~~

$$\begin{pmatrix} 30 & 40 & 50 \\ 45 & 60 & 75 \\ 60 & 80 & 95 \end{pmatrix} \Rightarrow 0 - 40 + 0 - 45 + 240 - 75 + 0 - 80 + 0 = 0$$

$$\textcircled{3} \begin{bmatrix} 30 & 45 & 60 \\ 45 & 60 & 80 \\ 60 & 80 & 100 \end{bmatrix} \quad L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\Rightarrow 0 - 45 + 0 - 45 + 2 \cdot 40 - 80 + 0 - 80 + 0$$

$$\Rightarrow -10. \quad (\text{Dark edge}).$$

$$\textcircled{4} \begin{bmatrix} 45 & 60 & 75 \\ 60 & 80 & 95 \\ 75 & 100 & 120 \end{bmatrix} \Rightarrow 320 - 95 + 0 - 100 + 0$$

$$\Rightarrow 320 - 120 - 95 - 100$$

$$= 320 - 220 - 95$$

$$= 320 - 315 = 5.$$

(00).

weak edge.

$$\textcircled{1} \rightarrow \begin{bmatrix} 20 & 30 & 40 \\ 30 & 45 & 60 \\ 45 & 60 & 80 \end{bmatrix}$$

$$\Rightarrow 4C - (T + B + L + R).$$

$$= 4 \times 45 - (30 + 60 + 30 + 60)$$

$$= 180 - 180 = 0.$$

$$\textcircled{2} \rightarrow \begin{bmatrix} 30 & 45 & 60 \\ 45 & 60 & 80 \\ 60 & 80 & 100 \end{bmatrix}$$

$$\Rightarrow 4C - (T + B + L + R)$$

$$\Rightarrow 4 \times 60 - (45 + 80 + 45 + 80)$$

$$\Rightarrow 240 - (170 + 80) = 250$$